Device-Related Thrombus after LAA Occlusion in Case 3: Image Assessment and Haemodynamic Risk

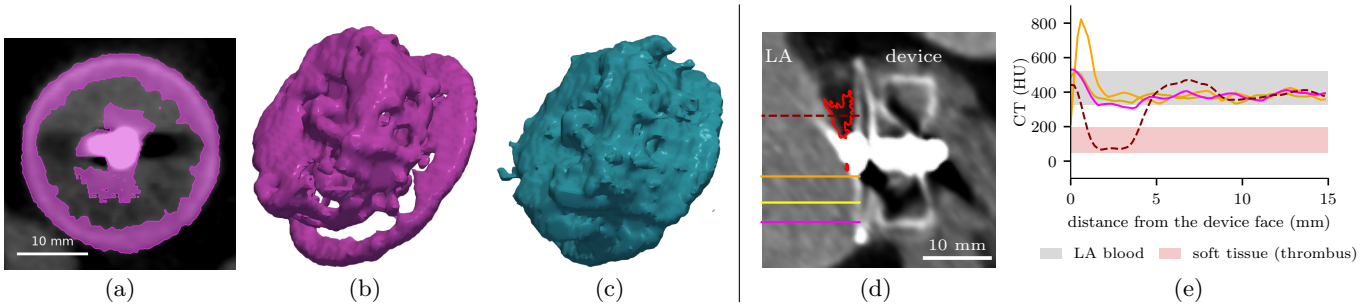
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Abstract

Is there a device-related thrombus (DRT)? No definite DRT is identified. Contrast-filled blood reaches the atrial face of the occluder, and the only low-density region lies beside the metal hub, with the shape of a streak artefact and a thickness below the 3 mm that defines a definite DRT. **Is the patient at risk of developing one?** Yes. A computational fluid dynamics (CFD) model of the left atrium (LA) shows that the blood next to the device is slower and less renewed than in the rest of the atrium, that the mixing vortices do not reach it, and that the surrounding wall has the lowest, most oscillating shear.



(a) (b) (c) (d) (e)

Figure 1: Device segmentation: (a) CT on the disc plane with the U-Net (nnInteractive) segmentation and (b) its 3D model; (c) manual segmentation, used in the analysis. Thrombus check: (d) CT along the device axis, thrombus window; red: the only regions below 250 HU in front of the device; (e) density along the lines in (d), from the device face into the LA (dashed: through the red region).

1 Introduction

Left atrial appendage occlusion (LAAO) prevents stroke in atrial fibrillation by closing the appendage with a device. A device-related thrombus (DRT) on the atrial face of the occluder is uncommon but increases the risk of stroke [1]. On contrast CT it appears as a hypoattenuating thickening (HAT) of the device surface [2]. Patient-specific CFD studies relate DRT to slow blood and to low, oscillating wall shear stress around the device [3, 4]. This report assesses Case 3 for an existing DRT and for the haemodynamic risk of developing one.

2 Methods

2.1 Segmentation and mesh

The contrast CT (diastolic phase) was segmented in 3D Slicer: the LA by manual thresholding of the contrast-filled blood, refined by painting and erasing within the threshold range, with the four pulmonary veins (PVs) up to their first branch and the device excluded. The Amulet-type device was segmented manually (Fig. 1c) and with a pretrained promptable 3D U-Net (nnInteractive [6]), prompted on the wires and the surrounding blood; its output was intersected with the local wire contrast of the CT and corrected in 3D Slicer. The U-Net recovers the rim and the hub, but the struts merge into thick bands (Fig. 1a,b). On the LA surface each PV was cut perpendicular to its trunk and extended by a 10 mm cylinder of the same area (diameters 9–18 mm); the mitral valve (MV) was cut on the annulus plane, aligned with the z axis and narrowed to a 5.3 cm^2 orifice. The volume was meshed with 272 000 tetrahedra (Gmsh).

2.2 Thrombus assessment

The CT was read in a thrombus window (−100 to 700 HU), in planes along and parallel to the device axis. Regions below 250 HU inside the LA, within 10 mm of the device face, were marked as candidates, and density profiles were taken from the face into the atrium. A DRT was defined as a non-enhancing layer of soft-tissue density on the atrial face; a layer thicker than 3 mm is a definite DRT [2, 7].

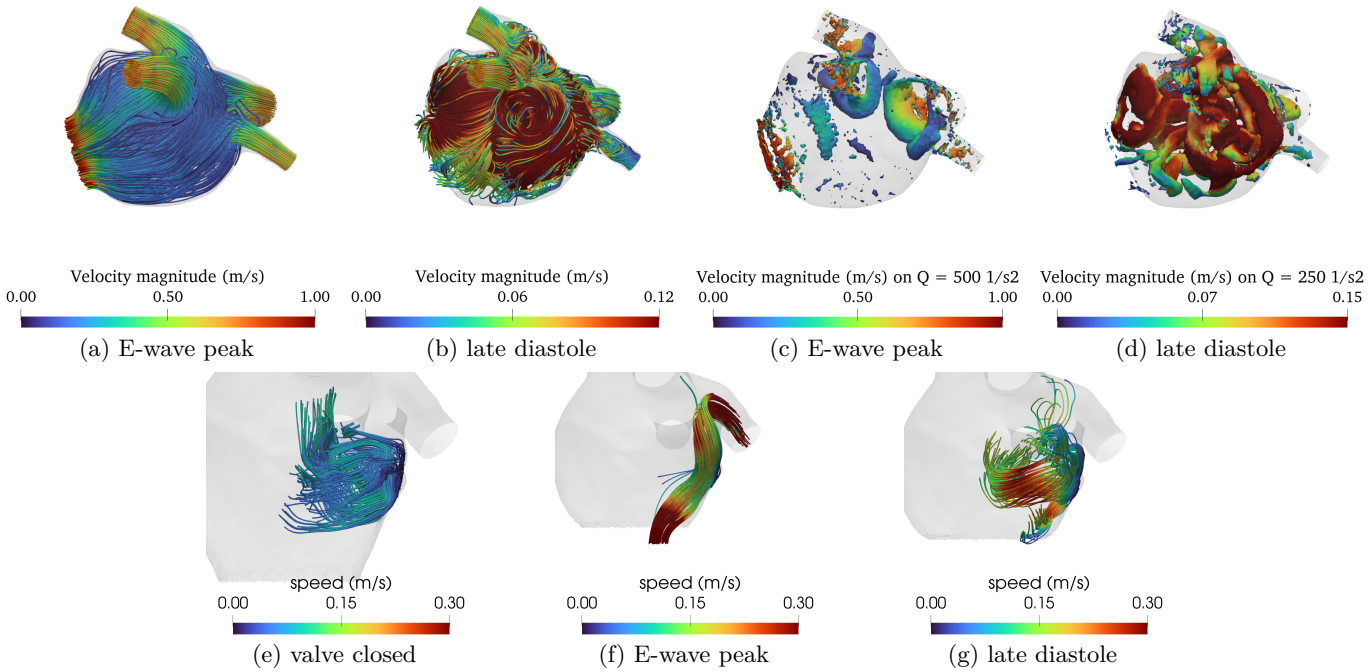
2.3 Flow simulation and analysis

Transient laminar flow of Newtonian blood ($\rho = 1060\,\mathrm{kg/m^3}$, $\mu = 3.5\,\mathrm{mPa\,s}$) was solved in Ansys Fluent, with the course pressure curve at the PVs and the course velocity curve at the MV (80 mL per beat, a normal output [8]) together with the longitudinal motion of the valve (dynamic mesh). Four beats were simulated ($\Delta t = 0.01\,\mathrm{s}$); all time steps met the residual criteria and the last beat, analysed, repeated the previous one. Streamlines and the Q-criterion, which separates swirl from wall shear, were examined in three phases (valve closed, E-wave peak, late diastole), and the mean speed and share of slow blood ($|v| < 5\,\mathrm{cm/s}$) were tracked within 5 and 10 mm of the device and in the whole LA. On the wall, the endothelial cell activation potential (ECAP, the oscillatory shear index divided by the time-averaged wall shear stress, TAWSS) marks regions of low and oscillating shear [4].

3 Results

3.1 Is there a thrombus?

In the CT, contrast-filled blood reaches the atrial face of the device over its whole height, without an interposed



(a) E-wave peak (b) late diastole (c) E-wave peak (d) late diastole

(e) valve closed (f) E-wave peak (g) late diastole

Figure 2: Flow in the last beat. (a, b) Streamlines from the pulmonary veins; (c, d) vortex structures (Q-criterion, $Q = 500$ and $250\,\mathrm{s}^{-2}$ [5]); (e–g) region in front of the device (not drawn, on the right), streamlines started 5 mm in front of it.

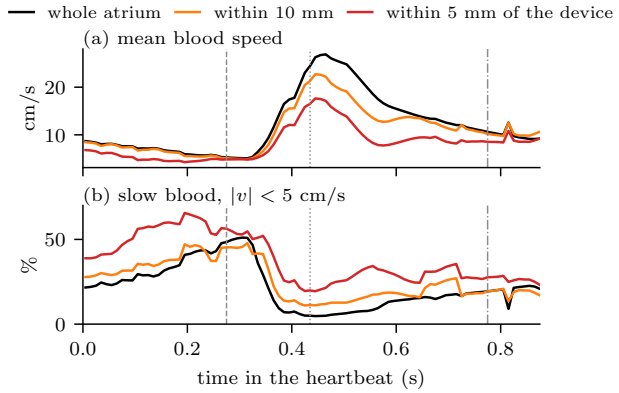
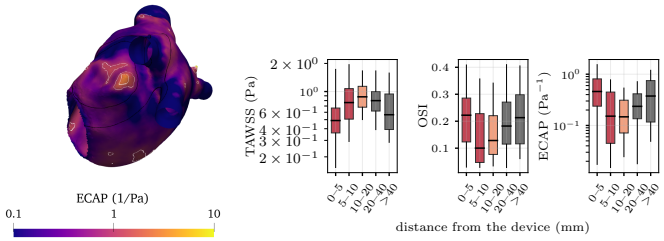


Figure 3: Blood near the device over the beat vs the whole LA. Vertical lines: phases of Fig. 2 (valve closed, E-wave peak, late diastole).



(a) (b)

Figure 4: (a) ECAP, device side (lines: 10 and 20 mm from the device). (b) Wall metrics by distance from the device.

layer of soft-tissue density (Fig. 1d). Below the hub, the density profiles pass directly from the device into blood density (Fig. 1e). The only candidate region is a thin dark band beside the metal hub, on both sides of it and in line with it. Its density is not uniform: soft-tissue values alternate with values below 0 HU, which a clot cannot have, and with bright bands, the typical pattern of a metal streak artefact; it is also thinner than 3 mm. No definite DRT is therefore present; a small low-grade HAT hidden by the artefact at the hub cannot be fully excluded on this single phase. Contrast is also visible in the appendage behind the device, which suggests incomplete sealing [7].

3.2 Flow near the device

At the E-wave peak, the flow from the four PVs converges into a jet towards the MV, with vortex rings at the vein inlets (Fig. 2a,c). In late diastole the flow slows down and a large vortex occupies the atrial body (Fig. 2b,d). The region in front of the device follows a different pattern. At the E-wave peak the main stream passes beside it without entering (Fig. 2f); with the valve closed and in late diastole the blood there is slow and turns on itself (Fig. 2e,g).

The time course confirms this (Fig. 3): within 5 mm of the device the mean speed is about one third lower than in the whole LA, most clearly during filling, and the share of slow blood is higher at all times, up to two thirds while the valve is closed. The vorticity near the device is high, but mostly as shear along the wall: the share of rotation-dominated blood ($Q > 0$) is 19 % within 5 mm against 33 % in the whole LA, and the vortex structures stay in the atrial body (Fig. 2c,d).

3.3 Wall shear stress

High ECAP appears at the entrance of the occluded appendage, inside the 10 mm line, and in small patches on the roof and the lower wall (Fig. 4a). Grouped by distance (Fig. 4b), the wall within 5 mm of the device has the lowest TAWSS and the highest OSI of the atrium, and a median ECAP about 1.7 times that of the remaining wall.

4 Discussion and conclusion

The device region combines slow, poorly renewed blood, few vortices and low, oscillating wall shear, the conditions associated with DRT in patient-specific studies [3, 4]. As the boundary conditions are generic and the device is not a solid in the flow domain, regions of the same atrium are compared rather than fixed thresholds.

Conclusion. No definite device-related thrombus is present on this CT, but **the patient is at risk of developing one**: the blood in front of the device is the slowest and least renewed of the atrium. Follow-up imaging is advisable.

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